Luxury Cosmetics Fraud Analysis — 2025

Python Data Analysis & Exploratory Data Analysis (EDA)

Project report following the supplied Python Data Analysis & EDA guideline.

1. Introduction and objective

This project explores transaction records for a luxury cosmetics business to describe transaction volumes, purchase amounts, product categories, payment methods, device types and records flagged by the source dataset. The goal is to demonstrate a reproducible analysis workflow and identify patterns that warrant further review.

2. Dataset description

The source CSV contains 2,133 rows and 18 columns. Each row represents a transaction record. Numeric fields include customer age, purchase amount, fraud flag and footfall count. Categorical fields include loyalty tier, location, store, product category, payment method and device type. Transaction date and time are available.

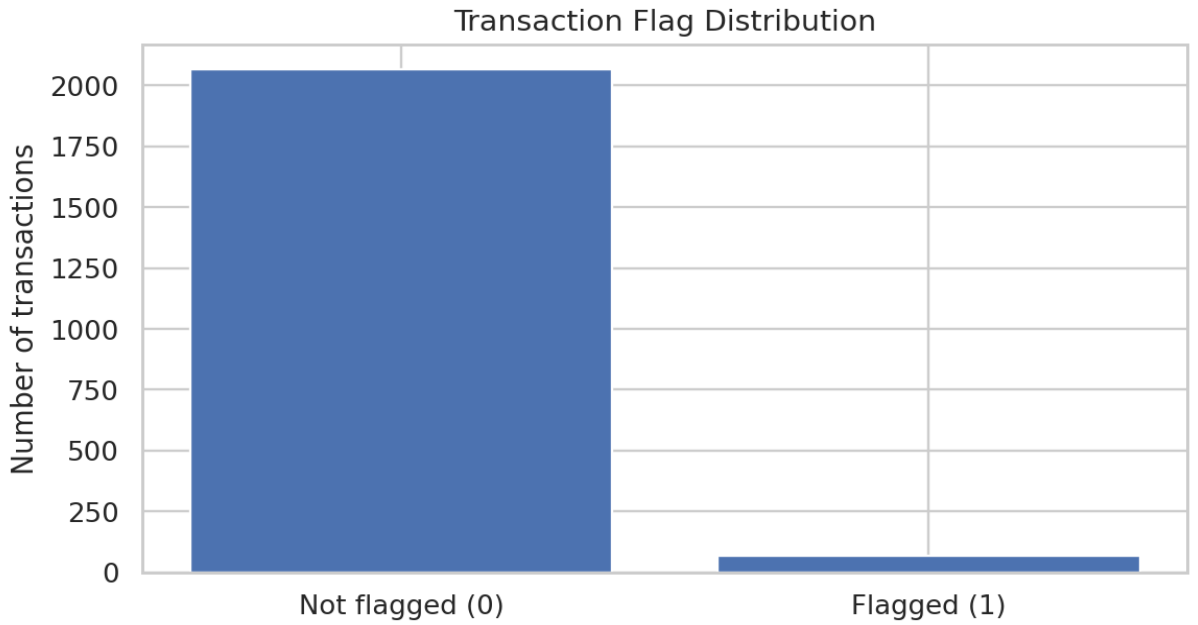
Metric	Result
Transactions	2,133
Columns	18
Flagged transactions	66
Flagged transaction rate	3.09%
Total purchase amount	372,451.82
Average purchase amount	174.61
Exact duplicate rows	0

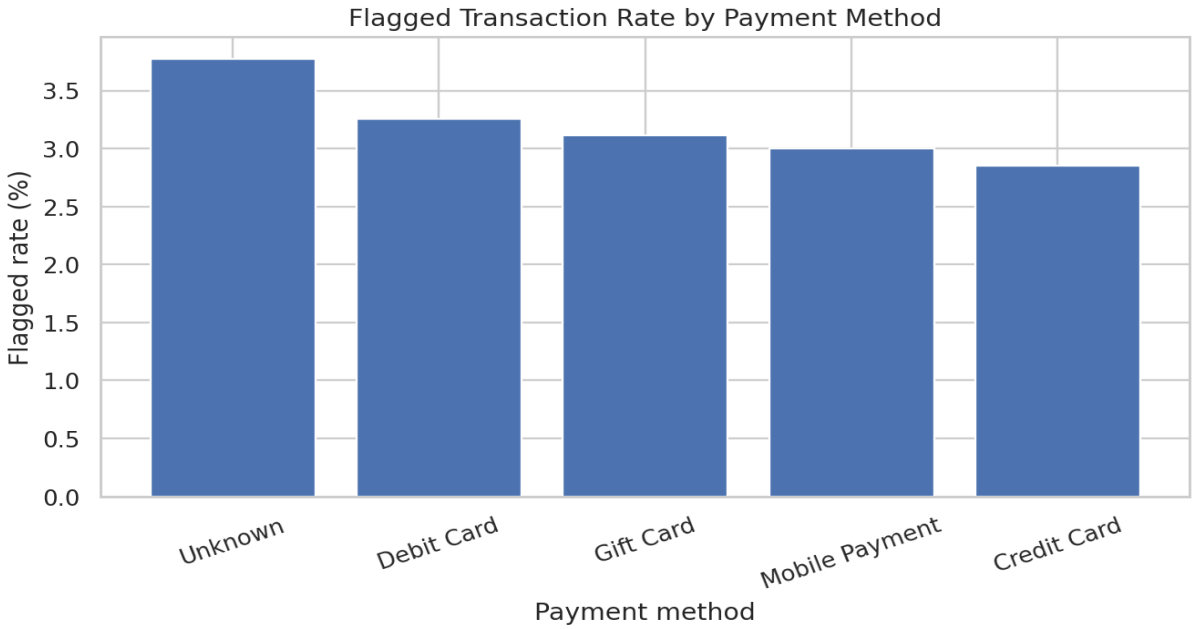
3. Data quality and cleaning

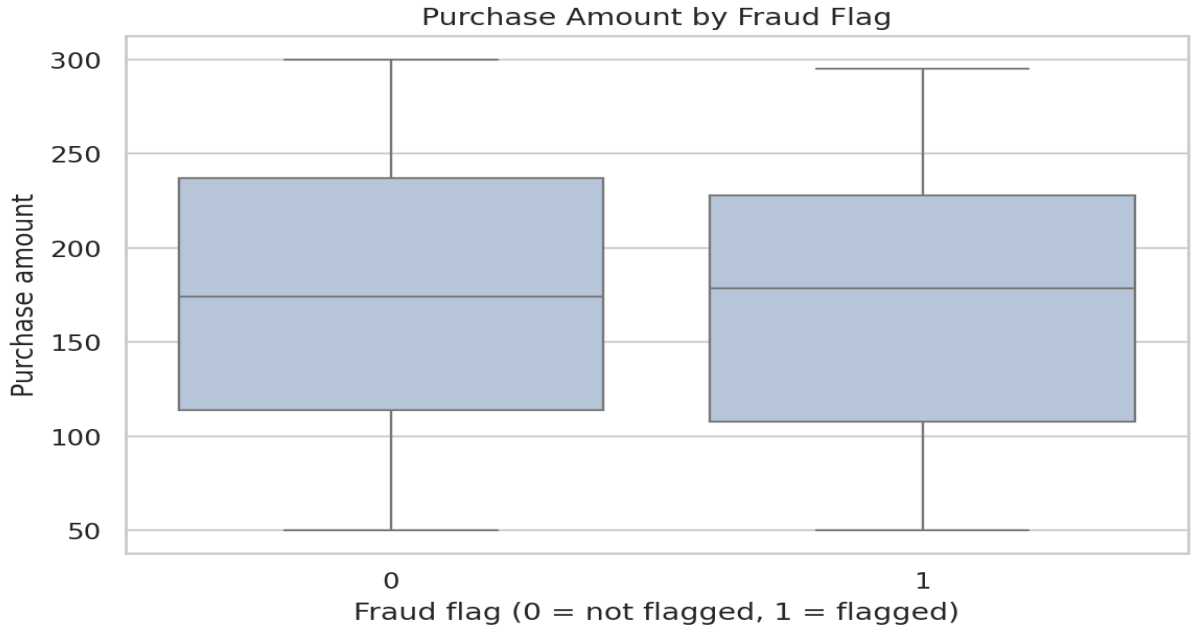
Missing values were found in Customer_Age (106), Customer_Loyalty_Tier (106) and Payment_Method (106). No exact full-row duplicates were found. Customer age was imputed with its median; missing loyalty/payment categories were labeled Unknown. Date/time and numeric fields were parsed to suitable types. Missing categorical values were not guessed, and no rows were removed merely for having unusual values.

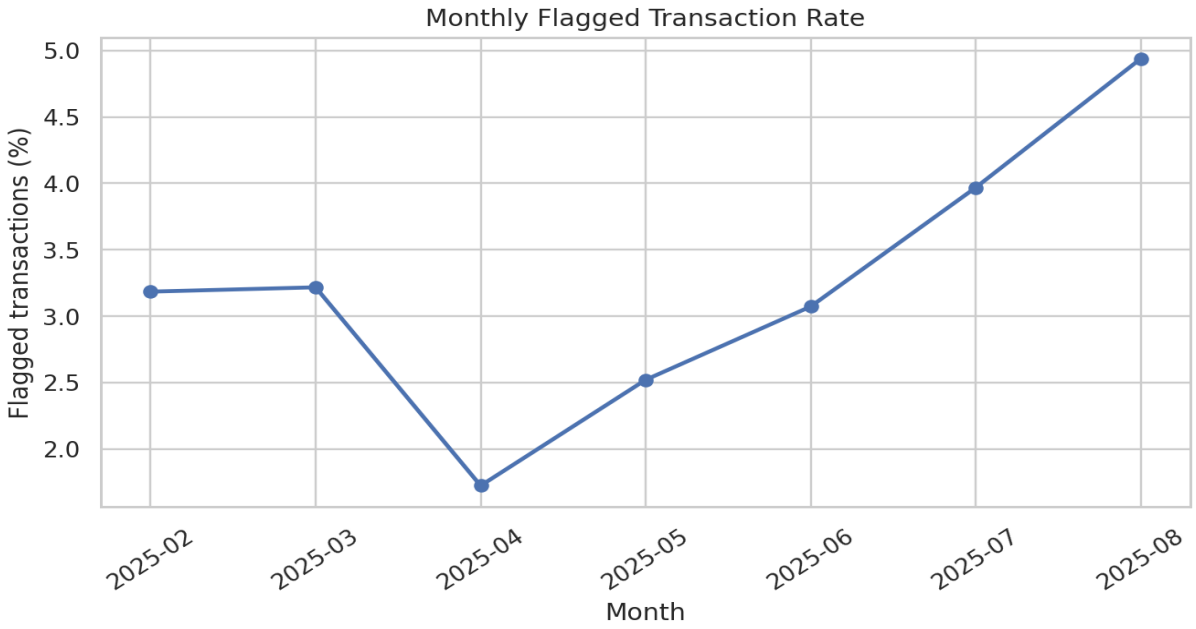
4. Exploratory analysis and visualizations

The charts below cover flag distribution, flagged rate by payment method, purchase amount comparisons, monthly trends, numeric correlations, category totals, and an additional payment-method × device-type risk matrix.


Transaction Flag Distribution
Number of transactions
2000
1750
1500
1250
1000
750
500
250
0
Not flagged (0)
Flagged (1)

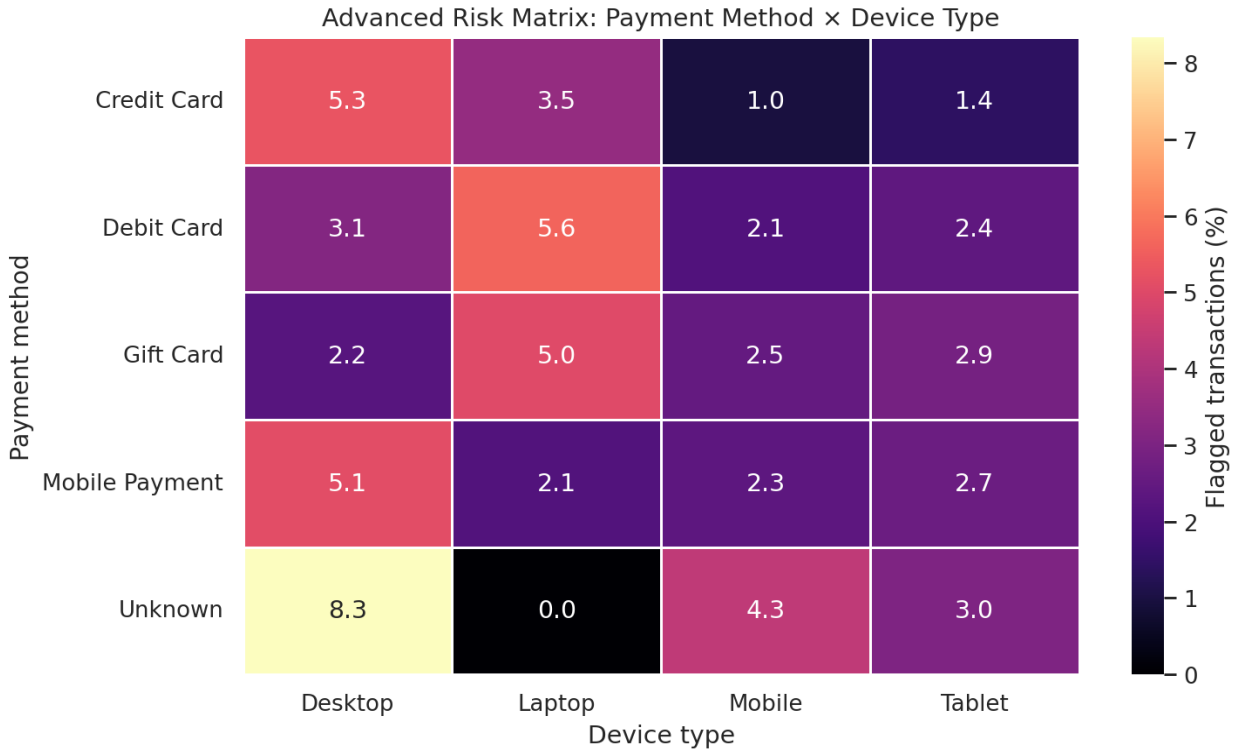

Flagged Transaction Rate by Payment Method
Flagged rate (%)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Unknown
Debit Card
Gift Card
Mobile Payment
Credit Card
Payment method


Purchase Amount by Fraud Flag
Purchase amount
300
250
200
150
100
50
0
1
Fraud flag (0 = not flagged, 1 = flagged)

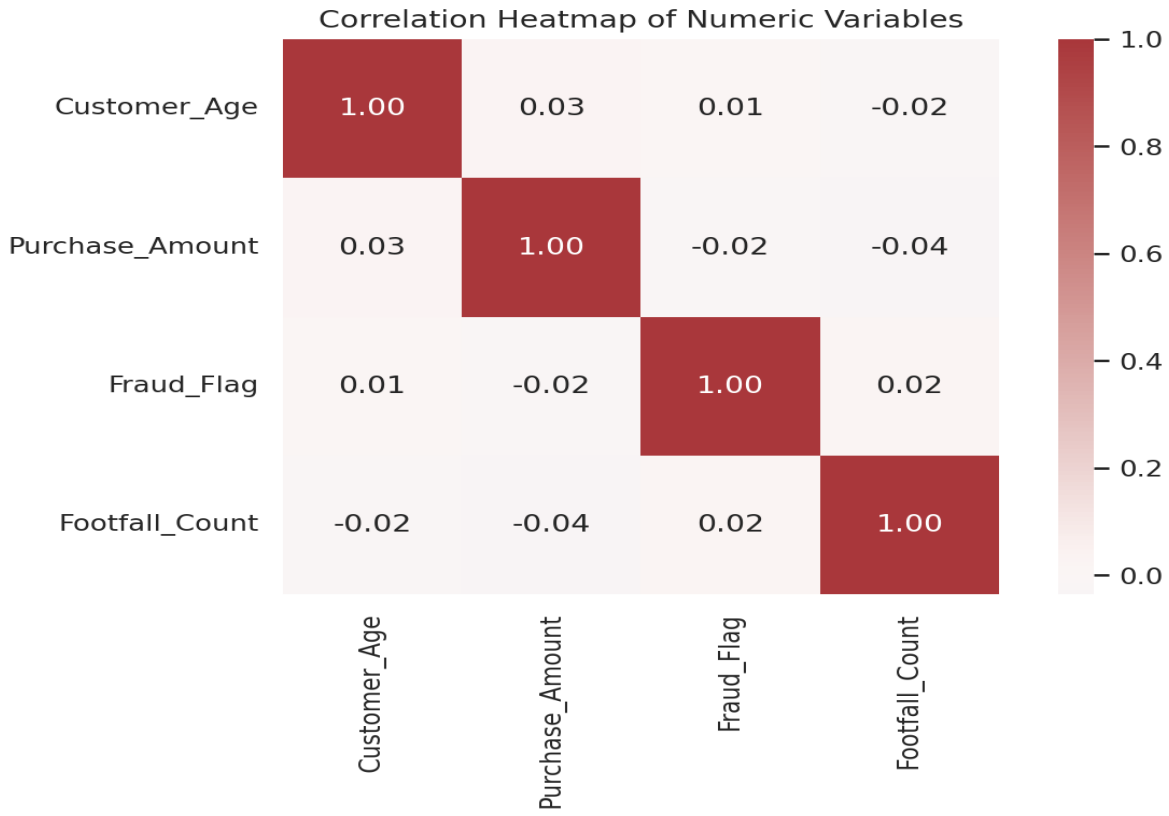

Monthly Flagged Transaction Rate
Flagged transactions (%)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
2025-02
2025-03
2025-04
2025-05
2025-06
2025-07
2025-08
Month

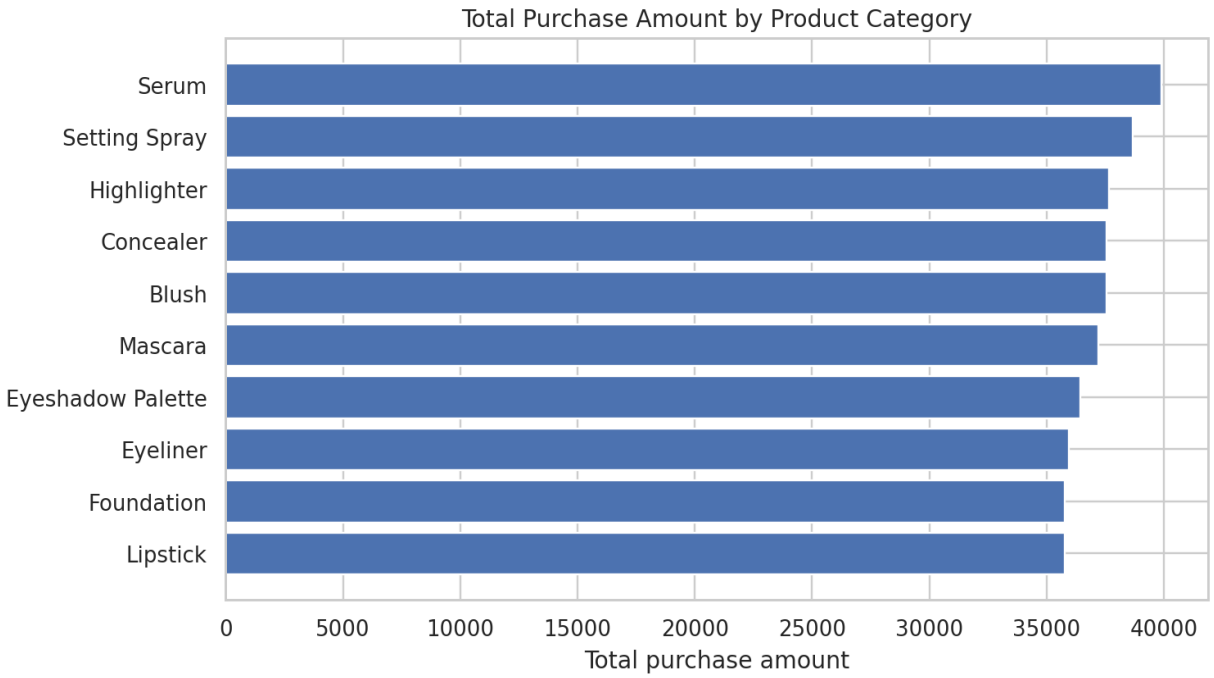
5. Advanced visualization

The payment × device heatmap is the extra graph: each cell shows the percentage of records flagged for that combination. Use both percentage and group size when interpreting it; small groups can produce volatile rates.



6. Correlation and category analysis





7. Key findings

• 66 of 2,133 transactions (3.09%) are flagged in the dataset.
• Total recorded purchase amount is 372,451.82; mean transaction amount is 174.61.
• 2025-08 has the highest observed monthly flagged rate (4.94%) in the months present; the monthly transaction counts vary.
• Serum has the highest total purchase amount among product categories (39,897.57).
• Missing payment method and loyalty tier are retained as Unknown for transparent grouping.

8. Conclusion and limitations

The analysis demonstrates data loading, inspection, cleaning, descriptive statistics, univariate/bivariate/multivariate analysis, grouping, sorting/filtering, pivot tables, crosstabs, NumPy calculations and Matplotlib/Seaborn visualizations. A flagged record is not proof of confirmed fraud; the dataset does not by itself establish the reason for a flag. Correlation does not prove causation. Findings should be validated against confirmed investigation outcomes and business rules before operational use.

Deliverables included: Jupyter notebook, cleaned CSV, data quality profile, chart images and this PDF report.